Reducing Regressivity in ML-Based Mass Appraisal: A Covariance-Penalty Layer for Assessor Workflows

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Abstract

Property taxes are a major source of public revenue in the U.S. local government. The processes that determine them are increasingly being produced by machine learning (ML) models. These models are usually optimized for predictive accuracy, which can leave issues such as *regressivity* in place or even amplify it: lower-value properties are over-assessed while higher-value properties are under-assessed. We introduce a covariance-based regressivity-control layer for standard valuation ML models. Starting from the observation that regressivity is, in assessor practice, a value-dependent pattern in assessment ratios, we add a soft penalty on a *proxy* of the empirical covariance between log-residuals and log-prices. We develop two forms of this penalty: a direct squared-covariance penalty and an implementation-friendly, sample-additive surrogate. The penalty plugs into standard ML models, such as gradient-boosting interfaces, through the usual custom gradient and Hessian routines, and closed-form solution for linear models. We apply the regressivity-correction method inside of the current pipeline of Cook County Assessor's Office (CCAO), which has roughly 350 thousand sales observations information and approximately 95 features of different nature. A single penalty hyperparameter recovers most of the *regressivity* gap. For example, for a representative surrogate-based configuration, the PRD deviation from its ideal baseline of 1 is reduced by approximately 55.3%, PRB improves by approximately 90.7% toward zero, and the absolute VEI error decreases by approximately 76.9%. Meanwhile, R^2 remains unchanged at 0.88, and COD and COV improve a slightly percentage, all of this relative to the unconstrained boosting baseline. We also formulate the hyperparameter and model selection process as a constrained calibration problem, selecting the model that maximizes predictive performance subject to explicit ratio-study tolerances. Overall, the complete workflow of the method adds a single regularization hyperparameter to existing assessor workflows, keeps the same preprocessing pipeline, feature set, and final diagnostics of accuracy and fairness. The method is designed as a minimal targeted correction layer inside of current valuation models, rather than a replacement for them.

1 Introduction

Property taxes are one of the largest sources of local public revenue in the United States, providing roughly 72% of local tax revenue and 47% of local own-source general revenue, or about $500 billion per year [6]. The values that generate those taxes are produced by *mass appraisal*: the estimation, as of a given date, of a large set of real estate values using common data, standardized methods, and statistical testing [23]. Assessor offices have increasingly turned to machine learning (ML) models such as gradient-boosted trees to carry out this task, motivated by their predictive performance on tabular data and their scalability to large parcel inventories [20, 7, 35, 37, 41]. In this setting, however, accuracy alone is insufficient. Assessments must also satisfy the operational constraints that assessors evaluate through standardized *ratio studies*: equity across similar properties (horizontal equity) and equity across value strata (vertical equity) [34, 38, 22, 24].

Vertical equity is typically the hardest constraint to satisfy in deployed systems. Assessments are *regressive* when lower-value properties are over-assessed and higher-value properties are under-assessed; such patterns are widespread across U.S. counties and carry substantial distributional consequences for taxpayers [31, 6, 21, 32]. Amornsiripanitch [3] estimates that owners of lower-valued properties face effective tax rates up to 50% higher than owners of higher-valued properties. Adopting a more flexible learner does not, by itself, resolve the problem. Squared-error objectives target conditional means, so when some price variation remains unexplained by the available features, the fit induces a systematic non-positive dependence between residuals and log-price; we make this precise in Section 3.1.1. The empirical counterpart is sharp in our CCAO study: moving from linear regression to a LightGBM baseline improves R^2, MAE, COD, and COV, yet worsens PRD, PRB, and VEI, pushing the model out of the assessor compliance region (Table 5).

This paper. We develop a regressivity-control layer that sits on top of the standard valuation learners already used by assessors. Starting from the sample covariance $\widehat{\mathrm{Cov}}(e, y)$ between log-residuals $e = \hat{y} - y$ and log-prices $y = \log P$, we add a soft penalty that discourages value-dependent trends in the fitted assessment ratios. We study two forms of this penalty:

- a *direct* squared-covariance penalty, which is the closest instantiation of the dependence-control idea;
- an *implementation-friendly* surrogate obtained as a separable upper bound on the squared covariance, which is sample-additive and therefore compatible with standard gradient-boosting interfaces such as LightGBM and XGBoost.

The direct penalty introduces dense curvature and is handled either through a diagonal Hessian approximation or through a custom leaf solver. The surrogate is separable by construction and is the variant we recommend for routine deployment. Both penalties add one scalar hyperparameter $\rho \geq 0$ and recover the unpenalized learner at $\rho = 0$.

We evaluate the method in a translated version of the Cook County Assessor's Office (CCAO) residential pipeline with 349,661 verified sales over 2016–2023, 95 features, and an expanding-window rolling-origin validation protocol. Relative to the LightGBM baseline, a single choice of ρ moves PRD from 1.085 to 1.038, PRB from -0.118 to -0.011, and VEI from -36.8% to -8.5%, with R^2 essentially unchanged at 0.882 and small improvements in COD and COV (Table 6). Because the penalty does not directly optimize ratio-study statistics, we also cast final model selection as a constrained calibration problem: choose the candidate that maximizes predictive performance subject to explicit ratio-study tolerances. Figure 1 shows the ratio–price pattern that the method is designed to flatten.

Contributions.

- *A covariance-based regressivity-control layer.* We formalize regressivity as a value-dependent pattern in assessment ratios and build a soft covariance penalty that can be added to standard regression losses. We develop both a direct squared-covariance penalty and an implementation-friendly separable surrogate. The surrogate is motivated by an upper bound on the squared covariance; dropping residual centering produces a sample-additive approximation that preserves the same value-weighted intuition and plugs into standard boosting APIs.
- *Implementation in standard learners.* We instantiate the penalties in regularized linear models and in second-order gradient-boosted trees. For the surrogate, we provide the exact gradient and Hessian that plug into LightGBM and XGBoost custom-objective interfaces. For the direct penalty, we provide both a diagonal approximation compatible with those interfaces and an exact Newton-style leaf solver (Appendix B).
- *Evidence in a real CCAO pipeline with constrained calibration.* Using the CCAO residential workflow, we show that a single penalty hyperparameter recovers most of the vertical-equity gap while preserving accuracy, and we cast final model selection as a constrained calibration problem that ranks candidates by predictive performance subject to explicit ratio-study tolerances.

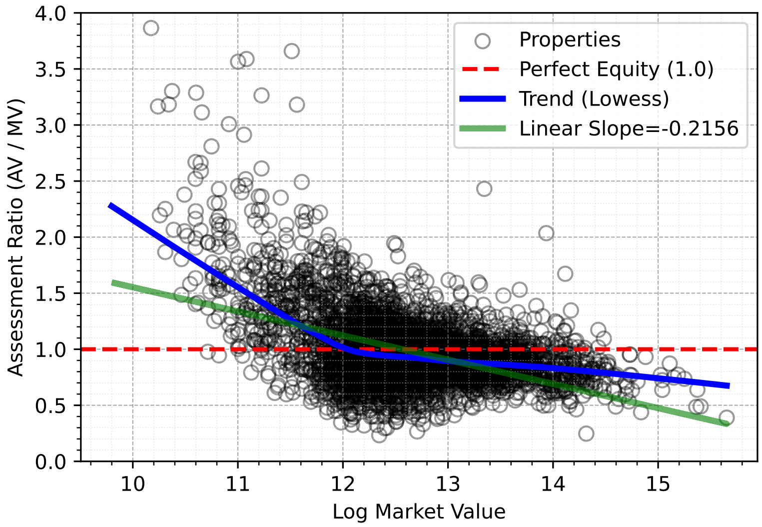


Figure 1: Regressivity in the baseline LightGBM fit on the 2023 Cook County test set. The assessment-to-sale ratio $r = \widehat{P}/P$ declines with property value: lower-value properties tend to be over-assessed ($r > 1$) and higher-value properties tend to be under-assessed ($r < 1$).

The paper is organized as follows. Section 2 fixes the assessment setting and the ratio-study diagnostics used throughout. Section 3 develops the covariance-based penalties and their implementation in linear models and boosted trees. Section 4 reports the CCAO evidence and the constrained calibration results. Section 5 concludes.

1.1 Related Work

The paper sits at the intersection of three literatures: machine learning for mass appraisal, equity and regressivity in assessor practice, and fair regression with continuous attributes. We cover each briefly and in the depth required to position the contribution.

ML in mass appraisal and assessor workflows. Recent reviews document the move from regularized linear models to tree ensembles and, more recently, neural architectures, with gradient-boosted trees now among the most common production learners [39, 35, 37, 41, 2]. Evaluation in this literature is still dominated by predictive criteria rather than equity as an explicit design target. Adoption in assessor offices is partial and uneven: Bidanset and Rakow [7] identify communication, technical capacity, and budget as the primary barriers, and the IAAO Artificial Intelligence Task Force [20] lists explainability, defensibility, administrative feasibility, and workflow integration as the conditions for real-world use. Any practical equity intervention therefore has to be compatible with these learners and layered onto the existing training pipeline.

Equity and regressivity in mass appraisal. Assessor standards decompose equity into horizontal equity, measured by the coefficient of dispersion (COD), and vertical equity, measured by the price-related differential (PRD), the coefficient of price-related bias (PRB), and the recently introduced vertical equity indicator (VEI) [22, 23, 24, 38, 16, 32]. IAAO Statistical Tools and Measures Task Force [21] recommend reporting a suite of these tests rather than relying on any single statistic, and McMillen and Singh [32] show that some regression-based regressivity detectors can be severely biased when applied to ML-generated assessments. Berry [6] documents that regressivity is widespread across U.S. counties and more consistent with assessors' data and model limitations than with pure measurement error, and Avenancio-León and Howard [4] document related tax-incidence effects. Corrections inside the valuation pipeline have been proposed via two-step spatial procedures [34], geographically weighted regression [8, 9], validation-time ratio studies [18, 19], and fairness-aware valuation models [11].

Fair regression and dependence-based regularization. Most early fair-ML work addresses classification, whereas property assessment is a continuous-prediction problem [12, 5]. Fair-regression methods treat fairness as an explicit objective through constrained empirical risk minimization [1, 17, 40], post-processing via optimal transport [14, 13], or explicit dependence constraints [25, 36]. The closest strand for our purposes is dependence-based regularization [30, 33, 27, 26, 10], which penalizes statistical dependence between model outputs and a continuous sensitive quantity. These regularizers require small changes to the base learner and are natural when the relevant inequity runs along a continuous variable rather than a demographic attribute, as is the case for assessment ratios and sale prices. For multi-objective selection we build on Martinez, Bertran, and Sapiro [29] and Little, Weylandt, and Allen [28].

The assessor literature defines regressivity as a value-dependent pattern in assessment ratios and addresses it mainly through diagnostics or post hoc corrections; the fair-ML literature provides training-time machinery to control such dependence but typically for demographic attributes rather than assessor-style vertical inequity. Our contribution is to recast the assessor-defined pattern as a covariance target on a standard valuation learner and to instantiate it inside the second-order boosting interfaces that assessor offices already use.

2 The Assessment Setting and the CCAO Workflow

This section fixes the assessor-facing ratio objects, the predictive notation, and the ratio-study diagnostics used throughout the paper. It also recasts regressivity as a covariance-based dependence pattern that links classical ratio-study diagnostics to the covariance penalties introduced in Section 3. Throughout, we anchor the exposition in the operational workflow of the Cook County Assessor's Office (CCAO), which also provides the empirical setting of Section 4.

The CCAO residential workflow. The CCAO produces residential valuations for roughly 1.5 million parcels on a triennial cycle using a LightGBM-based automated valuation model (AVM), and publishes the pipeline, feature set, and ratio-study diagnostics as open code. Sales are verified upstream and merged with 95 property, location, and neighborhood features; the learner is trained on verified arm's-length sales and evaluated on held-out years through ratio studies reported on the original price scale. Final candidate models are promoted only if they meet the office's internal ratio-study tolerances. Throughout the paper, we take this operational loop—train on historical verified sales, evaluate with price-scale ratios, screen with IAAO-style tolerances—as the deployment context that any regressivity correction must fit into. Section 4 describes our translation of this workflow into a Python-based implementation, the 2016–2022 training window, the 2023 held-out test year, and the 8-fold rolling-origin validation protocol.

2.1 Assessment Ratios and Equity Measures

For a fitted model and an evaluation sample S, let $\widehat{P}_i$ denote the predicted value of property i and P_i its observed sale price. The *assessment-to-sale ratio* is

$$r_i := \frac{\widehat{P}_i}{P_i}, \qquad \forall i \in S, \tag{1}$$

where dependence on the fitted model is suppressed when clear from context. We refer to r_i as the *assessment ratio*.

Ratio studies summarize assessment quality through statistics of $\{r_i\}_{i \in S}$ [24]. We use the standard IAAO equity diagnostics and the central statistics that measure the overall *assessment level*.

Remark. All equity diagnostics in this paper are computed on the price scale of $\widehat{P}_i$ and P_i, even though the underlying predictive models specified in later sections are trained on the log-prices scale.

Assessment level. The *assessment level* is summarized by central statistics of $\{r_i\}_{i \in S}$: the median, the mean, the value-weighted mean, and the standard deviation [24]. Let

$$m_S := \mathrm{med}\{r_i : i \in S\}$$

denote the median ratio,

$$\bar{r}(S) := \frac{1}{|S|} \sum_{i \in S} r_i$$

the mean ratio,

$$\overline{r_W}(S) := \frac{1}{\sum_{i \in S} P_i} \sum_{i \in S} r_i P_i$$

the value-weighted mean ratio, and

$$\sigma_S := \sqrt{\frac{1}{|S|-1} \sum_{i \in S} (r_i - \bar{r})^2}$$

the sample standard deviation of the assessment ratios. We omit the dependence on S when the context is clear.

Horizontal equity measures. We summarize *horizontal equity* primarily through the coefficient of dispersion (COD), which the IAAO describes as the average percentage deviation from the median ratio m_S [22, 24]:

$$\mathrm{COD}(S) := 100 \cdot \frac{1}{|S|} \sum_{i \in S} \frac{|r_i - m_S|}{m_S}. \tag{2}$$

Lower COD values indicate greater within-group uniformity. We also report the coefficient of variation (COV), defined as

$$\mathrm{COV}(S) := 100 \cdot \frac{\sigma_S}{\bar{r}}, \tag{3}$$

as a supplementary dispersion diagnostic. Both indicators summarize how dispersed the ratios are around a central tendency.

Vertical equity measures. We summarize vertical equity using standard *price-related diagnostics*. First, the price-related differential (PRD) is

$$\mathrm{PRD}(S) := \frac{\bar{r}}{\overline{r_W}}. \tag{4}$$

Under standard assessor interpretation, PRD values above 1.03 tend to indicate regressivity, while values below 0.98 tend to indicate progressivity [22, 24].

We also report the coefficient of price-related bias (PRB), defined in terms of a quantity referred to as the *market-value proxy*; [24] $\mathrm{MV}_i := \log_2\!\left(\frac{1}{2}\left(P_i + \frac{\widehat{P}_i}{m_S}\right)\right)$, designed to reduce measurement-error bias [24]. The PRB is then defined as

$$\mathrm{PRB}(S) := \mathrm{slope}\!\left(\frac{r_i - m_S}{m_S} \sim \mathrm{MV}_i\right). \tag{5}$$

That is, PRB is the slope of relative ratio deviations from the median against the market-value proxy. PRB values below -0.05 suggest regressivity, while values above 0.05 suggest progressivity [22, 24].

Finally, we report the vertical equity indicator (VEI), recently introduced in the IAAO 2025 exposure draft [24], defined as

$$\mathrm{VEI}(S) := 100 \cdot \frac{m_{S_{q_{10}}} - m_{S_{q_1}}}{m_S}, \tag{6}$$

where $m_{S_{q_{10}}}$ is the median ratio in the highest value decile of S (ranked by the proxy MV_i) and $m_{S_{q_1}}$ is the median ratio in the lowest value decile. VEI values below -10% suggest regressivity, while values above 10% suggest progressivity [24].

Table 1 summarizes the main assessor-facing diagnostics used later for empirical evaluation and constrained model selection.

Table 1: Summary of core assessment ratio statistics, desired targets, and interpretation.

Measure	**Target Value**	**Interpretation**
Central Statistics (Assessment Level)		
Median (m_S)	≈ 1	Outlier-robust measure of overall assessment level.
Mean ($\bar{r}$)	≈ 1	Arithmetic average.
Weighted Mean ($\overline{r_W}$)	≈ 1	Value-weighted average; denominator in the PRD.
Horizontal Equity (Uniformity)		
COD	$[5.0, 15.0]^*$	Average % deviation from the median. Lower values indicate better uniformity.
COV	–	Ratio of standard deviation to the mean. Lower values indicate less relative variability.
Vertical Equity (Regressivity)		
PRD	$[0.98, 1.03]$	Values > 1.03 indicate regressivity. Values $< .098$ indicate progressivity.
PRB	$[-0.05, 0.05]$	Values < -0.05 indicate regressivity. Values > 0.05 indicate progressivity.
VEI	$[-10\%, 10\%]$	Values < 0 indicate regressivity. Values > 0 indicate progressivity.

*Assessments with COD values below 5.0 are considered highly suspect. Acceptable upper limits vary by property type (up to 20.0 or 25.0 for rural/vacant) [24].

2.2 Regressivity as a Dependence Problem

The assessor literature describes regressivity as a price-related pattern in ratios: assessment ratios tend to fall as property value rises. Figure 2 shows this pattern as a negative trend between the ratios $\{r_i\}_{i\in S}$ and log-prices $\{y_i\}_{i\in S}$ with $y_i = \log P_i$. We plot against log-price because sale prices are highly right-skewed; log-price makes the trend comparable across value levels.

Because ratios and prices are continuous quantities, a natural first-order summary of their dependence is the empirical covariance:

$$\mathrm{Cov}(r, P) := \frac{1}{|S|}\sum_{i\in S}(r_i - \bar{r})(P_i - \bar{P}), \tag{7}$$

where $\bar{r}$ and $\bar{P}$ are the sample means of $\{r_i\}_{i\in S}$ and $\{P_i\}_{i\in S}$. When $\mathrm{Var}(r) > 0$ and $\mathrm{Var}(P) > 0$, the covariance, correlation, and simple-regression slope share a sign:

$$\mathrm{sign}(\mathrm{Cov}(r, P)) = \mathrm{sign}(\mathrm{Corr}(r, P)) = \mathrm{sign}(\mathrm{slope}(r \sim P)), \tag{8}$$

where $\mathrm{Corr}(r, P) := \frac{\mathrm{Cov}(r,P)}{\sqrt{\mathrm{Var}(r)\mathrm{Var}(P)}}$, $\mathrm{slope}(r \sim P) := \frac{\mathrm{Cov}(r,P)}{\mathrm{Var}(P)}$, and $\mathrm{sign}(\cdot) \in \{-1, 0, 1\}$ is the signum function. Thus $\mathrm{Cov}(r, P) < 0$ captures a first-order price-related negative pattern, which we associate with regressivity. The covariance is also directionally aligned with the standard price-related differential (PRD) used in assessor practice [24], as formalized in Proposition 1.

Proposition 1. *Let $P > 0$ and $r = \widehat{P}/P > 0$ almost surely, with $\mathbb{E}[r] < \infty$, $\mathbb{E}[P] < \infty$, and $\mathbb{E}[rP] < \infty$. Then the population covariance and the population* $\mathrm{PRD} = \frac{\mathbb{E}[r]\mathbb{E}[P]}{\mathbb{E}[rP]}$ *satisfy*

$$\mathrm{sign}(\mathit{Cov}(r, P)) = \mathrm{sign}(1 - \mathrm{PRD}). \tag{9}$$

Proof. Since $\mathrm{Cov}(r, P) = \mathbb{E}[rP] - \mathbb{E}[r]\mathbb{E}[P]$, we have

$$1 - \mathrm{PRD} = 1 - \frac{\mathbb{E}[r]\mathbb{E}[P]}{\mathbb{E}[rP]} = \frac{\mathbb{E}[rP] - \mathbb{E}[r]\mathbb{E}[P]}{\mathbb{E}[rP]} = \frac{\mathrm{Cov}(r, P)}{\mathbb{E}[rP]}.$$

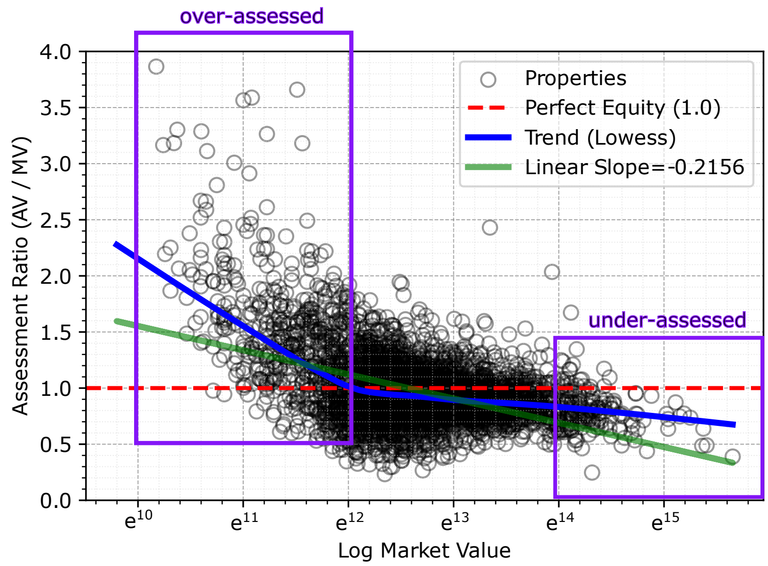


Figure 2: Illustration of regressive behavior in mass appraisal. Assessment-to-sale ratios r tend to decrease with property value P: lower-value properties are more likely to be over-assessed ($r > 1$), while higher-value properties are more likely to be under-assessed ($r < 1$).

Because $r > 0$ and $P > 0$ almost surely, $\mathbb{E}[rP] > 0$. Therefore, $1 - \text{PRD}$ and $\text{Cov}(r, P)$ have the same sign. $\square$

Proposition 1 shows that the covariance $\text{Cov}(r, P)$ is directionally consistent with the PRD, which links the ratio-study view of regressivity to the dependence-based formulation developed in the next sections.

2.3 Mass-Appraisal Prediction Setting

We observe a sample of n sold properties $\mathcal{D}_n := \{(x_i, P_i, t_i)\}_{i=1}^n$, where $x_i \in \mathcal{X} \subseteq \mathbb{R}^p$ is a feature vector (e.g., square footage, bedroom count, distance to transit), $P_i \in \mathbb{R}_{++}$ is the verified sale price, and t_i is the sale date. Because prices are positive and right-skewed, we work on the log-price scale, $y_i := \log P_i$.

Let Θ be a parameter space and $f_\theta : \mathcal{X} \to \mathbb{R}$, $\theta \in \Theta$, a candidate predictor of log-price (e.g., $f_\theta(x_i) = \theta^\top x_i$ in linear models). Each $i \in [n]$ receives a log-scale prediction $\hat{y}_i := f_\theta(x_i)$, with corresponding price-scale prediction $\widehat{P}_i := \exp(\hat{y}_i)$.

The baseline tension. The empirical problem this paper addresses can be stated already at this level. The CCAO workflow moved from linear regression to a LightGBM AVM to exploit the predictive gains of tree ensembles on tabular real-estate data. On the translated 2023 CCAO test set (Section 4, Table 5), LightGBM lifts R^2 from 0.762 to 0.883 and cuts MAE from \$86,257 to \$72,064 relative to linear regression, and it also improves horizontal-equity diagnostics (COD from 25.0% to 23.1%, COV from 36.8% to 35.4%). Over the same comparison, the vertical-equity diagnostics move in the opposite direction: PRD rises from 1.039 to 1.085, PRB falls from −0.011 to −0.118, and VEI drops from −13.5% to −36.8%. The median ratio slides from 1.001 to 0.944. In operational terms, the stronger predictor exits the assessor compliance region on exactly the dimension the office most cares about. The covariance penalty introduced next is designed to recover vertical-equity behavior without giving up this predictive ground, and the remainder of the paper is organized around that single tension.

Log-scale residuals as a ratio proxy. Models are trained on the log-price scale, so we define the *log-scale residual* $e_i := \hat{y}_i - y_i, \quad i \in [n]$. By construction, e_i is the log-assessment ratio:

$$e_i = \hat{y}_i - y_i = \log \widehat{P}_i - \log P_i = \log\left(\frac{\widehat{P}_i}{P_i}\right) = \log r_i, \qquad \forall i \in S. \tag{10}$$

Because $\operatorname{sign}(e_i) = \operatorname{sign}(r_i - 1)$, the sign of e preserves the interpretation of over- and under-assessment. Identity (10) also yields the log-scale counterpart of the price-related dependence of Section 2.2: $\operatorname{Cov}(e, y) = \operatorname{Cov}(\log r, \log P)$. This is the quantity the method of Section 3 penalizes during training.

3 Covariance-Based Penalization in Standard Learners

3.1 General Baseline Learning Setting

Let $\mathcal{H} := \{f_\theta : \theta \in \Theta\}$ denote the hypothesis class induced by a parameter space Θ, and let $\mathcal{D}_n$ denote a training sample of size n with $y_i = \log P_i$. The empirical loss is

$$L(f, \mathcal{D}_n) := \frac{1}{n} \sum_{i=1}^{n} \ell(f(x_i), y_i),$$

where ℓ is a prediction-error loss on the log-price scale. The canonical choice in assessor workflows is the squared residual loss, $\ell(f(x_i), y_i) = (f(x_i) - y_i)^2 = e_i^2$, which we use throughout; the general notation is kept so that the approach extends to alternative losses.

Baseline learner. The *baseline* learner solves the empirical risk minimization problem

$$f_{\text{base}} \in \arg\min_{f \in \mathcal{H}} \left\{ L(f, \mathcal{D}_n) + \Omega(f) \right\}, \tag{11}$$

where $\Omega : \mathcal{H} \to \mathbb{R}$ is an optional complexity regularizer on Θ (e.g., ℓ_2 regularization in ridge regression). We refer to f_{base} as the baseline predictor and to (11) as the baseline learning problem.

3.1.1 Why regressive patterns are expected in squared-loss learning

Squared-loss learning connects directly to the dependence view of regressivity developed in Section 2.2. In that setting the baseline predictor f_{base} targets the conditional mean $\mathbb{E}[y \mid x]$, and any residual conditional variation in log-price that cannot be explained from features generates a systematic non-positive linear dependence between log-residuals and log-prices. Proposition 2 makes this precise.

Proposition 2 (Population covariance non-positivity under squared-error prediction)**.** *Let $Y := \log P$ be the log-sale price, let X denote the observed feature vector, and assume $\mathbb{E}[Y^2] < \infty$ and $\mathbb{E}[P^2] < \infty$. Let $f^\star(X) = \mathbb{E}[Y \mid X]$ be the Bayes predictor under squared-error loss. Define the associated population residual $e^\star := f^\star(X) - Y$.*

Let $g : \mathbb{R} \to \mathbb{R}$ be a measurable nondecreasing function such that $\mathbb{E}[g(Y)^2] < \infty$. Then

$$\mathit{Cov}(e^\star, g(Y)) = -\mathbb{E}\big[\mathit{Cov}(Y, g(Y) \mid X)\big] \le 0. \tag{12}$$

Proof. By the law of total covariance,

$$\operatorname{Cov}(e^\star, g(Y)) = \mathbb{E}\big[\operatorname{Cov}(e^\star, g(Y) \mid X)\big] + \operatorname{Cov}\big(\mathbb{E}[e^\star \mid X],\, \mathbb{E}[g(Y) \mid X]\big).$$

Since

$$\mathbb{E}[e^\star \mid X] = \mathbb{E}[\mathbb{E}[Y \mid X] - Y \mid X] = 0,$$

the second term is zero. Moreover, conditional on X, the quantity $\mathbb{E}[Y \mid X]$ is constant, so

$$\operatorname{Cov}(e^\star, g(Y) \mid X) = \operatorname{Cov}(\mathbb{E}[Y \mid X] - Y,\ g(Y) \mid X) = -\operatorname{Cov}(Y, g(Y) \mid X).$$

Hence

$$\mathrm{Cov}(e^\star, g(Y)) = -\mathbb{E}\big[\mathrm{Cov}(Y, g(Y) \mid X)\big].$$

Finally, since g is nondecreasing, Chebyshev's covariance inequality applied under the conditional law of $Y \mid X$ yields

$$\mathrm{Cov}(Y, g(Y) \mid X) \geq 0 \qquad \text{a.s.}$$

Therefore,

$$\mathrm{Cov}(e^\star, g(Y)) \leq 0.$$

$\square$

Taking $g(Y) = Y$ or $g(Y) = \exp(Y)$ in (12) shows that the non-positive dependence holds both between log-residuals and log-prices and between log-residuals and price-scale values. This is a population statement and does not imply that every finite-sample squared-loss fit will be regressive under the assessor-side diagnostics of Section 2.1. It does motivate augmenting (11) with a dependence-aware correction when mitigating this behavior is a goal.

3.2 General Constrained and Penalized Setting

Sections 2.2 and 3.1.1 show that ML-based mass appraisal can induce value-dependent distortions in assessment ratios, both empirically and at the population level. We now introduce a general correction class that constrains or penalizes a sample summary of that dependence.

Let $\Psi(f, \mathcal{D}_n)$ denote a sample functional that quantifies value-dependent distortions in the fitted log-residuals or, equivalently, in the fitted log-assessment ratios. We can reformulate (11) as the constrained problem

$$\begin{aligned} \min_{f \in \mathcal{H}} \quad & L(f, \mathcal{D}_n) + \Omega(f) \\ \text{s.t.} \quad & \Psi(f, \mathcal{D}_n) \leq \tau, \end{aligned} \tag{13}$$

for a tolerance $\tau \geq 0$. The hard constraint is conceptually useful because τ directly parameterizes the maximum dependence allowed at training. For estimation and tuning across model classes, we use its soft-constrained Lagrangian counterpart, the *penalized learning problem*

$$f_\rho \in \arg\min_{f \in \mathcal{H}} \left\{ L(f, \mathcal{D}_n) + \Omega(f) + \rho\, \Psi(f, \mathcal{D}_n) \right\}, \tag{14}$$

where $\rho \geq 0$ controls the tradeoff between predictive fit and regressivity correction, and $\rho = 0$ recovers the baseline (11). Formulation (14) captures the structure of our approach: augment a standard predictive learner with a term that discourages systematic value-dependent distortions in the fitted assessments. Sections 3.3 and 3.4 specialize Ψ to covariance-based choices and then implement those choices in linear models and second-order tree boosting.

3.3 Covariance-Based Penalization

We now instantiate (14) with concrete covariance-based choices of Ψ. By (10), $e_i(f) = \log r_i(f)$, so penalizing dependence between the fitted log-residuals $e(f) := (e_1(f), \ldots, e_n(f))$ and the log-prices $y := (y_1, \ldots, y_n)$ is equivalent to penalizing dependence between fitted log-assessment ratios and log-sale prices.

3.3.1 Squared covariance penalty

From Section 2.2, the empirical covariance $\widehat{\mathrm{Cov}}(e(f), y)$ is proportional to the slope of the least-squares regression of log-residuals on log-prices, $\mathrm{slope}(e(f) \sim y) = \widehat{\mathrm{Cov}}(e(f), y)/\widehat{\mathrm{Var}}(y)$, and $\widehat{\mathrm{Var}}(y)$ does not depend on f. Controlling this covariance therefore controls the first-order value-dependent trend in fitted assessment ratios. The most symmetric choice penalizes the squared covariance:

$$\Psi^{\mathrm{cov}^2}(f) := \frac{1}{2}\, \widehat{\mathrm{Cov}}(e(f), y)^2. \tag{15}$$

Substituting (15) into (14) yields

$$f_\rho^{\mathrm{cov}^2} \in \arg\min_{f\in\mathcal{H}} \left\{ L(f) + \Omega(f) + \frac{\rho}{2}\,\widehat{\mathrm{Cov}}(e(f), y)^2 \right\}. \tag{16}$$

This is the most direct instantiation of the dependence-control idea developed above: it suppresses regressive and progressive first-order trends symmetrically, and is appropriate when the target is to drive the covariance toward zero.

3.3.2 A separable surrogate penalty

The squared covariance penalty has two limitations. As a first-order summary, covariance can miss non-monotone dependence patterns in which local positive and negative trends cancel. Moreover, $\widehat{\mathrm{Cov}}(e(f), y)^2$ is non-separable across observations, which complicates implementation in optimizers that rely on per-observation gradients and Hessians, most notably the second-order tree-boosting interfaces that dominate mass-appraisal practice. We therefore upper-bound the squared covariance by a quantity that is additive over observations:

$$\widehat{\mathrm{Cov}}_n(e(f), y)^2 = \left(\frac{1}{n}\sum_{i=1}^{n} \big(e_i(f) - \bar{e}(f)\big)\big(y_i - \bar{y}\big) \right)^2 \le \frac{1}{n}\sum_{i=1}^{n} \big(e_i(f) - \bar{e}(f)\big)^2 \big(y_i - \bar{y}\big)^2, \tag{17}$$

where the inequality follows from Jensen's inequality. The bound still couples observations through $\bar{e}(f)$, so we drop the residual centering and use the separable surrogate

$$\Psi^{\mathrm{surr}}(f) := \frac{1}{n}\sum_{i=1}^{n} e_i(f)^2 \big(y_i - \bar{y}\big)^2. \tag{18}$$

Substituting (18) into (14) gives the surrogate-penalized problem

$$f_\rho^{\mathrm{surr}} \in \arg\min_{f\in\mathcal{H}} \left\{ L(f) + \Omega(f) + \rho\, \Psi^{\mathrm{surr}}(f) \right\}. \tag{19}$$

Ψ^{surr} is best understood as a covariance-motivated, observation-additive regularizer rather than as exact control of $\widehat{\mathrm{Cov}}_n(e(f), y)^2$. Two features make it attractive for assessor deployment. First, it is fully additive across observations, so it plugs into standard gradient-boosting custom-objective interfaces without modifying the leaf-splitting logic. Second, it weights each squared residual by the squared price deviation $(y_i - \bar{y})^2$, so residual distortions far from the center of the value distribution are penalized most heavily. This is the property that drives the strongest empirical behavior in Section 4.

Remark. By suppressing the residual average $\bar{e}(f)$, the surrogate penalizes the distance between $e_i(f)$ and zero instead of the distance to the average log-residual. On the original price scale, the analogous quantity is the squared deviation $(r_i - 1)^2$. Because the surrogate weights by $(y_i - \bar{y})^2$ in log space, it treats equal multiplicative deviations around the geometric center of the price distribution symmetrically, which is the scale on which the ratio–price trend is naturally read.

Methodological roles of the two penalties. The two penalties are not interchangeable. The direct penalty in (15) is the closest realization of the dependence-control target: it suppresses first-order linear dependence between log-ratios and log-prices symmetrically and in either direction. The surrogate in (18) sacrifices exact covariance control in exchange for three properties that matter for deployment: it is sample-additive, so it composes cleanly with standard gradient-boosting custom-objective interfaces; it cannot be zeroed out by local cancellation between positive and negative residuals, so it responds to pointwise distortions rather than only to aggregate covariance; and its per-observation weight $(y_i - \bar{y})^2$ is fixed before training, so tuning the penalty amounts to a one-dimensional line search on ρ. In the empirical study of Section 4, the surrogate delivers the strongest combined behavior on the CCAO workflow, and we recommend it as the default for routine deployment. The direct penalty is retained in the main text because it exposes the object that the surrogate approximates and because it admits the exact leaf-wise Newton implementation of Appendix B.

3.4 Model-Specific Implementations

We now instantiate the covariance-based penalties of Section 3.3 in two model classes already used in mass-appraisal practice: linear models and second-order tree boosting. Linear models offer a transparent analytical benchmark that exposes the algebraic effect of each penalty. Second-order tree boosting covers the high-performing nonlinear learners (XGBoost, LightGBM) that dominate recent assessor pipelines. The implementation choices highlighted here are what make the method drop-in compatible with standard training infrastructure: the surrogate maps directly onto per-observation custom objectives, and the squared covariance reduces to a manageable rank-one correction inside the per-round tree optimization.

3.4.1 Linear Covariance-Penalized Models

Consider the linear hypothesis class

$$\mathcal{H}_{\text{lin}} := \{f_\theta : x \mapsto x^\top \theta \ : \ \theta \in \Theta \subseteq \mathbb{R}^p\},$$

and let $X \in \mathbb{R}^{n \times p}$ denote the feature matrix. For a given coefficient vector $\theta \in \Theta$, we have $f_\theta(x_i) = x_i^\top \theta$, and $e := X\theta - y$. For notation simplicity, let us define

$$H := I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \qquad c := Hy = y - \bar{y}\,\mathbf{1}, \qquad D := \operatorname{diag}(c_1^2, \dots, c_n^2),$$

so that $\widehat{\operatorname{Cov}}(e, y) = \frac{1}{n} e^\top c$.

Throughout this subsection, we take the baseline loss to be

$$L(f_\theta) = \frac{1}{2n}\|X\theta - y\|_2^2.$$

Squared covariance penalty in linear models. For the symmetric penalty

$$\Psi^{\text{cov}^2}(f_\theta) = \frac{1}{2}\,\widehat{\operatorname{Cov}}_n(e, y)^2 = \frac{1}{2n^2}\big(e^\top c\big)^2,$$

the penalized estimator solves

$$\theta_\rho^{\text{cov}^2} \in \arg\min_{\theta \in \Theta} \left\{ \frac{1}{2n}\|X\theta - y\|_2^2 + \frac{\rho}{2n^2}\big((X\theta - y)^\top c\big)^2 \right\}. \tag{20}$$

Define

$$a := X^\top c, \qquad b := y^\top c = c^\top c.$$

Then the first-order optimality condition for (20) is

$$\left(\frac{1}{n}X^\top X + \frac{\rho}{n^2} a a^\top\right)\theta = \frac{1}{n}X^\top y + \frac{\rho}{n^2} b a.$$

Hence, whenever the displayed inverse exists,

$$\theta_\rho^{\text{cov}^2} = \left(\frac{1}{n}X^\top X + \frac{\rho}{n^2} a a^\top\right)^{-1} \left(\frac{1}{n}X^\top y + \frac{\rho}{n^2} b a\right). \tag{21}$$

This shows that the squared covariance penalty induces a rank-one modification of the usual normal equations.

Surrogate penalty in linear models. For the separable surrogate

$$\Psi^{\mathrm{surr}}(f_\theta) = \frac{1}{n}\sum_{i=1}^{n} e_i^2 c_i^2 = \frac{1}{n} e^\top D\, e,$$

the penalized problem is

$$\theta_\rho^{\mathrm{surr}} \in \arg\min_{\theta\in\Theta} \left\{ \frac{1}{2n}\|X\theta - y\|_2^2 + \frac{\rho}{n} e^\top D\, e \right\}. \tag{22}$$

The corresponding optimality condition is

$$\left(\frac{1}{n}X^\top X + \frac{2\rho}{n}X^\top D X\right)\theta = \frac{1}{n}X^\top y + \frac{2\rho}{n}X^\top D y,$$

and therefore, whenever the displayed inverse exists,

$$\theta_\rho^{\mathrm{surr}} = \left(\frac{1}{n}X^\top X + \frac{2\rho}{n}X^\top D X\right)^{-1}\left(\frac{1}{n}X^\top y + \frac{2\rho}{n}X^\top D y\right). \tag{23}$$

Extension to regularized linear models. The same constructions extend to penalized linear estimators

$$\min_{\theta\in\Theta} \{L(f_\theta) + \Omega(\theta) + \rho\, \Psi(f_\theta)\}$$

whenever Ω is a convex, differentiable regularizer. Non-differentiable cases such as the LASSO remain convex but lose the closed form and must be solved numerically. In both regimes, the covariance penalty integrates into standard regularized linear regression as a small modification of the normal equations.

3.4.2 Covariance-Penalized Second-Order Tree Boosting

We next consider second-order additive tree boosting under squared loss, as used in gradient-boosted tree systems such as XGBoost and LightGBM. Let $f^{(m-1)}$ denote the current prediction function at boosting iteration m, and write

$$f_i := f^{(m-1)}(x_i), \qquad e_i := f_i - y_i, \qquad c_i := y_i - \bar{y}, \qquad z := \widehat{\mathrm{Cov}}_n(e, y) = \frac{1}{n}\sum_{i=1}^{n} e_i c_i.$$

At step m, one adds a regression tree t to the current predictor, so that

$$f^{(m)}(x) = f^{(m-1)}(x) + \nu\, t(x),$$

where $\nu \in (0, 1]$ is the learning rate.

Adding dependence penalties into the boosting objective. For standard second-order boosting, the objective is approximated locally by

$$\widetilde{\mathcal{J}}(t) = \sum_{i=1}^{n}\left(g_i\, t_i + \frac{1}{2}h_i\, t_i^2\right) + \Omega(t), \tag{24}$$

where $t_i := t(x_i)$, $\Omega(t)$ is the tree-complexity penalty, and

$$g_i := \frac{\partial}{\partial f_i}\left(L(f) + \rho\, \Psi(f)\right), \qquad h_i := \frac{\partial^2}{\partial f_i^2}\left(L(f) + \rho\, \Psi(f)\right)$$

are evaluated at the current predictions. Under squared loss,

$$\frac{\partial L(f)}{\partial f_i} = \frac{e_i}{n}, \qquad \frac{\partial^2 L(f)}{\partial f_i^2} = \frac{1}{n}.$$

The covariance-based penalties modify these quantities as follows.

Squared covariance penalty in boosting objective. For $\Psi^{\text{cov}^2}(f) = \frac{1}{2}z^2$, the gradient is

$$g_i^{\text{cov}^2} = \frac{e_i}{n} + \frac{\rho}{n} z\, c_i. \tag{25}$$

The exact Hessian of the penalized objective with respect to the prediction vector $f = (f_1, \ldots, f_n)$ is

$$\nabla_f^2 \big(L(f) + \rho\, \Psi^{\text{cov}^2}(f) \big) = \frac{1}{n} I + \frac{\rho}{n^2} cc^\top. \tag{26}$$

Unlike the separable boosting objectives used in standard libraries, (26) is dense: each observation is coupled to every other observation through the rank-one term $cc^\top$. Consequently, there are two natural implementation routes.

The first is a practical approximation that keeps only the diagonal terms,

$$h_i^{\text{cov}^2,\text{diag}} = \frac{1}{n} + \frac{\rho}{n^2} c_i^2. \tag{27}$$

This produces a custom objective compatible with standard second-order boosting APIs, at the price of ignoring the cross-observation curvature induced by the covariance penalty.

The second is an exact leaf-wise implementation for a fixed tree structure. Suppose a candidate tree t has leaves $I_1, \ldots, I_J$ and leaf weights $w = (w_1, \ldots, w_J)$, so that $t_i = w_j$ for $i \in I_j$. Let

$$n_j := |I_j|, \qquad b_j := \sum_{i \in I_j} e_i, \qquad s_j := \sum_{i \in I_j} c_i,$$

and let $s = (s_1, \ldots, s_J)^\top$. If $\Omega(t) = \gamma J + \frac{\lambda}{2}\|w\|_2^2$, then the exact local objective for the squared covariance penalty can be written as

$$\widetilde{\mathcal{J}}_{\text{cov}^2}(w) = g_T^\top w + \frac{1}{2} w^\top A_T w + \gamma J + \text{const}, \tag{28}$$

where

$$g_{T,j} = \frac{b_j}{n} + \frac{\rho}{n} z\, s_j, \qquad A_T = \operatorname{diag}\Big(\frac{n_1}{n} + \lambda, \ldots, \frac{n_J}{n} + \lambda \Big) + \frac{\rho}{n^2} ss^\top.$$

Hence the optimal leaf weights satisfy

$$A_T w^\star = -g_T, \qquad w^\star = -A_T^{-1} g_T, \tag{29}$$

whenever A_T is invertible. The associated gain for the fixed structure T is

$$\text{Gain}(T) = \frac{1}{2} g_T^\top A_T^{-1} g_T - \gamma J. \tag{30}$$

This exact route preserves the full curvature of the covariance penalty, but requires a custom tree-construction procedure rather than the usual observation-wise Hessian interface.

Surrogate penalty as a sample-additive boosting objective. For the surrogate

$$\Psi^{\text{surr}}(f) = \frac{1}{n} \sum_{i=1}^n e_i^2 c_i^2,$$

the gradient and Hessian remain observation-wise separable:

$$g_i^{\text{surr}} = \frac{e_i}{n} + \frac{2\rho}{n} e_i c_i^2, \tag{31}$$

$$h_i^{\text{surr}} = \frac{1}{n} + \frac{2\rho}{n} c_i^2. \tag{32}$$

Accordingly, the surrogate penalty is particularly well suited to off-the-shelf second-order boosting implementations: once the weights c_i^2 are precomputed, the penalized objective can be supplied through standard custom gradient and Hessian routines.

Implementation summary. The squared covariance penalty is the most direct implementation of the covariance-control idea, but it introduces dense curvature (Hessian); in practice, it can be handled either through a diagonal Hessian approximation in standard boosting libraries (XGBoost or LightGBM), or through an exact custom leaf solver of the form (29). The surrogate penalty is fully sample-additive, and is therefore the most convenient to implement through standard observation-wise gradient and Hessian interfaces.

The implementation details in this section show that the covariance-based framework is compatible with both linear estimators and high-performing nonlinear learners. This model-specific flexibility is important for the practical goal of the paper: regressivity correction can be layered onto standard predictive tools already used in assessment workflows, rather than requiring a separate valuation architecture.

Remark. For completeness, Appendix B provides pseudocode for an exact Newton-style tree-boosting implementation of the squared covariance penalty.

4 Empirical Study: Cook County Assessor's Office

4.1 Scope, Pipeline, and Evaluation Design

Our empirical study is based on the residential valuation workflow developed by the Cook County Assessor's Office (CCAO) [15]. The scope is restricted to *residential* properties in Cook County, Illinois, covering single-family homes, townhomes, small multi-family buildings, and condominiums. For the experiments, we follow the data preparation, preprocessing, and feature-engineering structure of the `model-res-avm` codebase, and in particular the workflow corresponding to the 2025 assessment cycle. The original workflow is largely implemented in R, whereas our experiments are run in Python 3.10.14 as a translation of that pipeline. The temporal split is chosen to mimic a realistic assessor setting: 2016–2022 are used for model development, 2023 is used as an out-of-time test year, and 2024 is reserved as the current assessment year for later deployment analysis. Unless otherwise stated, the empirical claims in this section are based on the validation and test components of this translated pipeline.

Because the implementation is a translation rather than a direct execution of the original production code, the results below should be interpreted as evidence within the translated CCAO workflow. This distinction matters for reproducibility, but it does not change the main empirical question of the section: whether a covariance-based correction can improve the accuracy–equity tradeoff within a realistic assessor pipeline.

4.2 Data Description

Sample construction. The dataset consists of verified residential property sales in Cook County, drawn from the CCAO's 2025 assessment-cycle data. Following the CCAO pipeline, we exclude multicard PINs (properties with multiple assessment cards) and sales flagged as outliers by the office's verification process. After these filters, the modeling-ready sample contains approximately 349,661 observations. Table 2 reports the temporal splits used throughout the study.

Table 2: Temporal data splits and sample sizes.

Split	**Period**	**Observations**
Train / Validation	Jan. 2016 – Dec. 2022	314,694
Test (held-out)	Jan. 2023 – Dec. 2023	34,967
Total (pre-2024)		*349,661*

Predictors. The model uses 95 predictor features, of which 23 are treated as categorical. Table 3 summarizes the features by category. All features are constructed upstream by the CCAO pipeline; our implementation consumes the already-built feature set without additional feature engineering for LightGBM models.

Table 3: Summary of predictor features by category.

Category	Count	Examples
Property characteristics	25	Year built, bedrooms, baths, building/land sqft
Proximity	20	Distance to transit, parks, hospitals, highways
ACS Census (5-year)	16	Median income, education, poverty rate
Location	8	Lat/lon, flood factor, walkability, school districts
Time	8	Sale year, quarter, day-of-week, post-COVID indicator
Parcel shape	6	Vertex count, edge length SD, bounding rectangle ratios
Other / identifiers	12	Township, neighborhood code, tax bill rate
Total	**95**	

Validation protocol. Within the 2016–2022 development sample, we construct 8 rolling-origin validation folds using an expanding-window protocol. Each fold uses a fixed validation block of approximately 31,469 observations (10% of the development sample), while the training window expands chronologically from approximately 35,054 to 255,337 observations across folds. Validation windows step forward in 9-month increments, covering the period from late 2016 through mid-2022. Table 4 reports the fold structure.

Table 4: Rolling-origin validation fold structure. Training windows expand chronologically; validation blocks are of fixed size.

Fold	Train start	Train end	Val. start	Val. end	Train size
0	Jan. 2016	Sep. 2016	Oct. 2016	Jul. 2017	35,054
1	Jan. 2016	Jul. 2017	Jul. 2017	Mar. 2018	66,523
2	Jan. 2016	Mar. 2018	Mar. 2018	Nov. 2018	97,992
3	Jan. 2016	Nov. 2018	Nov. 2018	Aug. 2019	129,461
4	Jan. 2016	Aug. 2019	Aug. 2019	May 2020	160,930
5	Jan. 2016	May 2020	May 2020	Jan. 2021	192,399
6	Jan. 2016	Jan. 2021	Jan. 2021	Aug. 2021	223,868
7	Jan. 2016	Aug. 2021	Aug. 2021	May 2022	255,337

Validation block size: ≈31,469 observations per fold.

4.3 Baseline Tension: Linear Versus Nonlinear Models

We first compare two baseline learners used in the CCAO workflow: linear regression and LightGBM. Table 5 reports held-out test-set results. LightGBM substantially improves predictive fit and horizontal-equity diagnostics relative to the linear model, raising R^2 and lowering MAE, COD, and COV. At the same time, PRD, PRB, and VEI move sharply in the regressive direction. The assessment-level columns confirm the pattern: Linear Regression attains a median ratio near unity (1.001), whereas the LightGBM median ratio drops to 0.944, indicating systematic under-assessment at the typical property value.

This is the central empirical motivation for the paper. A more flexible learner captures market structure better but induces a stronger value-dependent pattern in the assessment ratios. The proposed correction is designed to preserve the predictive advantages of LightGBM while reducing this regressivity.

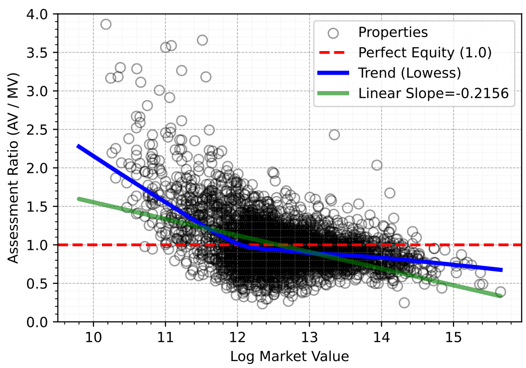
4.4 Effect of Covariance Penalization

The regressive pattern is visible in the ratio–value plots of Figure 3. Under the baseline LightGBM, the fitted ratios exhibit a strong negative trend with linear slope ≈ -0.2156 and ratio–price correlation ≈ -0.2666. The squared covariance penalty moves these to ≈ -0.1263 and ≈ -0.1224; the surrogate penalty moves

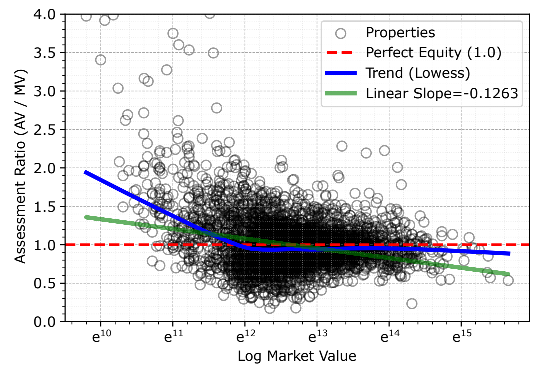
Table 5: Baseline comparison on the 2023 test set. Lower is better for MAE, COD, and COV. For PRD, median r, and mean r, values closer to one are preferred; for PRB and VEI, values closer to zero are preferred.

Model	R^2	MAE	Med. r	Mean r	COD	COV	VEI	PRD	PRB
Linear Regression	0.762	$86,257	1.001	1.051	25.013%	36.8%	-13.474%	1.039	-0.011
LGBMRegressor	0.883	$72,064	0.944	1.021	23.106%	35.4%	-36.816%	1.085	-0.118

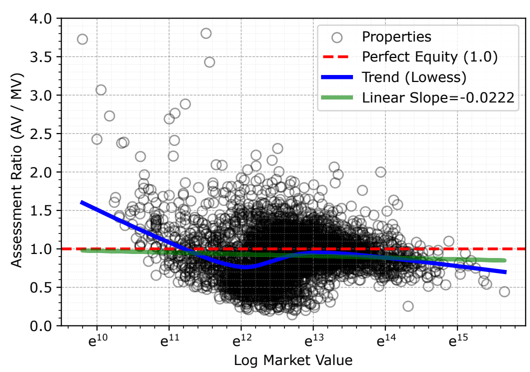
them further toward zero, to ≈ -0.0222 and ≈ -0.0239. These slope and correlation summaries are not the assessor-facing evaluation criteria, but they confirm that the training-time penalty acts in the intended direction: fitted assessment ratios become less systematically related to property value as ρ grows.



(a) Baseline LightGBM



(b) Squared-covariance-penalized LightGBM



(c) Surrogate-penalized LightGBM

Figure 3: Mechanism view of covariance penalization. Relative to the baseline model, the penalized learners visibly flatten the ratio–value trend. In the current experiment, the linear slope moves from approximately -0.2156 to -0.1263, while the ratio–price correlation moves from approximately -0.2666 to -0.1224 for the squared-covariance penalization. For the surrogate penalization, the linear slope moves to -0.0222 and the ratio–price correlation to -0.0239.

4.5 Penalized LightGBM Performance on the Test Set

Table 6 reports held-out test results for representative penalized LightGBM configurations. Relative to the unconstrained LightGBM baseline, the penalized models move vertical-equity diagnostics sharply toward assessor-preferred ranges. The best squared-covariance and surrogate configurations remain competitive on predictive accuracy and horizontal equity, and in several cases improve them.

A representative configuration is the surrogate-penalized LightGBM at $\rho = 6.572$. Relative to the LightGBM baseline, R^2 stays at 0.882, COD improves from 23.106 to 22.806, and COV improves from 35.4% to 33.4%. MAE rises modestly from $72,064 to $74,428, a cost of roughly 3.3% in predictive error. Over the same configuration, VEI moves from -36.8% to -8.5%, PRD from 1.085 to 1.038, and PRB from -0.118 to -0.011. The PRB and VEI reach IAAO-consistent ranges; PRD is close to, but still outside, the conventional $[0.98, 1.03]$ tolerance. Larger ρ values (e.g., $\rho = 26.519$) push PRD inside the tolerance at the cost of a slightly larger drop in R^2, illustrating the equity–accuracy tradeoff the penalty is designed to expose.

Table 6: Held-out test-set performance for representative penalized LightGBM configurations. The penalty parameter ρ is penalty-specific and should not be compared across different penalty definitions.

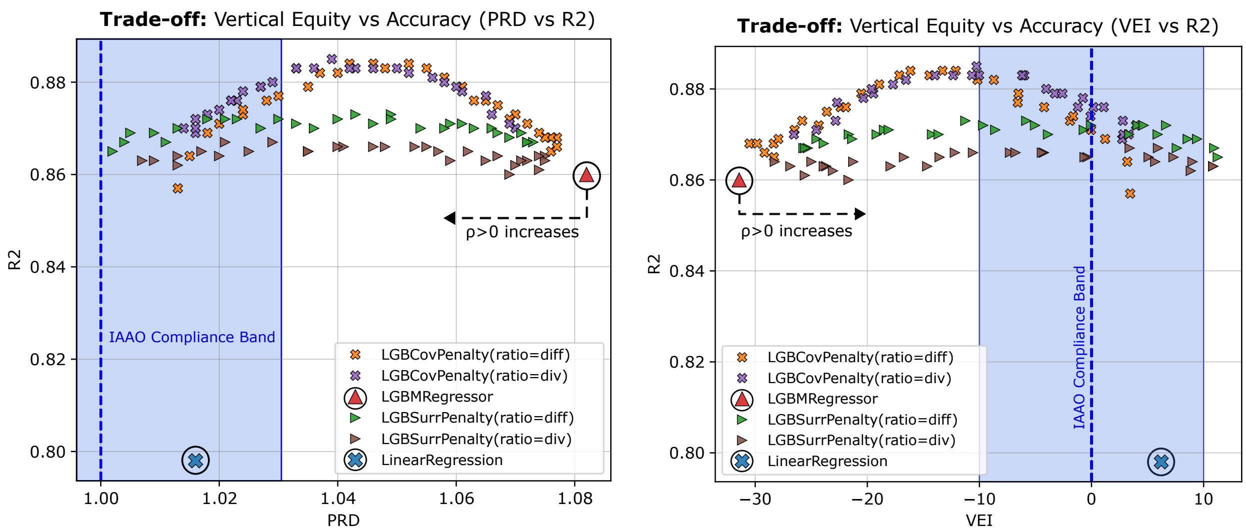
Model Family	ρ	R^2	MAE	Med. r	Mean r	COD	COV	VEI	PRD	PRB
Linear Regression	–	0.762	$86,256.879	1.001	1.051	25.013%	36.825%	-13.474%	1.039	-0.011
LGBMRegressor	–	0.883	$72,063.853	0.944	1.021	23.106%	35.398%	-36.816%	1.085	-0.118
LGBSurrogatePenalty	0.100	0.881	$73,271.882	0.936	1.011	23.125%	35.236%	-36.367%	1.084	-0.115
LGBSurrogatePenalty	0.189	0.881	$73,208.883	0.936	1.010	23.042%	35.048%	-33.608%	1.080	-0.107
LGBSurrogatePenalty	0.404	0.882	$73,110.847	0.937	1.006	22.924%	34.782%	-29.878%	1.075	-0.095
LGBSurrogatePenalty	0.761	0.883	$72,974.386	0.937	1.001	22.749%	34.189%	-23.789%	1.067	-0.076
LGBSurrogatePenalty	1.628	0.882	$73,233.014	0.938	0.994	22.645%	33.664%	-16.431%	1.057	-0.052
LGBSurrogatePenalty	3.070	0.881	$73,663.987	0.937	0.986	22.667%	33.620%	-12.630%	1.048	-0.032
LGBSurrogatePenalty	6.572	0.882	$74,428.114	0.937	0.976	22.806%	33.386%	-8.545%	1.038	-0.011
LGBSurrogatePenalty	12.390	0.879	$75,846.355	0.935	0.968	23.249%	33.684%	-7.446%	1.032	0.003
LGBSurrogatePenalty	26.519	0.879	$77,113.783	0.933	0.960	23.719%	33.959%	-4.484%	1.024	0.020
LGBSurrogatePenalty	50.000	0.878	$78,155.478	0.932	0.956	24.092%	34.336%	-4.883%	1.021	0.027
LGBCovPenalty	0.100	0.879	$73,459.227	0.937	1.014	23.210%	35.551%	-38.368%	1.087	-0.122
LGBCovPenalty	0.189	0.880	$73,311.803	0.939	1.015	23.221%	35.443%	-37.671%	1.086	-0.119
LGBCovPenalty	0.404	0.882	$72,947.453	0.941	1.017	23.224%	35.432%	-36.501%	1.083	-0.115
LGBCovPenalty	0.761	0.883	$72,607.948	0.944	1.019	23.157%	35.246%	-34.017%	1.078	-0.106
LGBCovPenalty	1.628	0.886	$72,198.279	0.949	1.023	23.176%	35.086%	-30.427%	1.071	-0.096
LGBCovPenalty	3.070	0.884	$72,293.558	0.955	1.027	23.216%	35.053%	-26.850%	1.065	-0.085
LGBCovPenalty	6.572	0.881	$72,871.805	0.961	1.030	23.282%	35.040%	-20.525%	1.055	-0.068
LGBCovPenalty	12.390	0.878	$73,449.207	0.963	1.030	23.261%	34.757%	-16.050%	1.048	-0.057
LGBCovPenalty	26.519	0.870	$74,741.535	0.963	1.028	23.310%	34.542%	-13.572%	1.041	-0.046
LGBCovPenalty	50.000	0.866	$75,574.792	0.960	1.023	23.357%	34.522%	-11.105%	1.037	-0.039

4.6 Accuracy–Equity Tradeoffs

Increasing ρ traces out a nontrivial accuracy–equity frontier rather than a steep linear tradeoff. In PRD–R^2 space, the penalized models move away from the unconstrained LightGBM point toward the assessor compliance region while remaining close to the upper edge of the accuracy frontier. The same pattern holds in VEI–R^2 space. This is the empirical justification for treating calibration as a constrained model-selection problem: once multiple diagnostics matter simultaneously, the operational question is not which model ranks best on a single score, but which model delivers the strongest predictive performance among those that satisfy explicit ratio-study tolerances.

4.7 Multi-Objective Model Selection Across Folds

Once the implemented model families are fit (with or without covariance-based penalization), a final candidate must be chosen for deployment. Because different configurations achieve different tradeoffs across predictive error, vertical-equity diagnostics, and dispersion, calibration is naturally formulated as a constrained multi-objective selection problem over the candidate set.



(a) PRD versus R^2

(b) VEI versus R^2

Figure 4: Accuracy–equity tradeoffs induced by covariance penalization on the test set. As ρ increases, the penalized models move away from the unconstrained LightGBM point toward assessor-preferred vertical-equity ranges while remaining near the upper part of the accuracy frontier.

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4.7.1 Multi-Objective Model-Selection Setting

Let $\mathcal{V} = \{1, \ldots, V\}$ index the validation folds (or rolling-origin validation windows), and let V_v denote the validation sample associated with fold $v \in \mathcal{V}$. Let $\mathcal{C}$ denote a finite set of candidate model configurations. Each configuration $c \in \mathcal{C}$ specifies (i) a model family from the implemented classes, (ii) a penalty choice in $\Psi \in \{0, \Psi^{\mathrm{cov}^2}, \Psi^{\mathrm{surr}}\}$, where $\Psi = 0$ denotes the unpenalized baseline, and (iii) all associated tuning parameters, including the penalty level $\rho \geq 0$ when applicable.

For each fold $v \in \mathcal{V}$, fitting configuration c on the corresponding training portion yields a predictor $f_{c,v}$. For each observation $i \in V_v$, let $\hat{y}_i^v(c) := f_{c,v}(x_i^v)$, $\hat{P}_i^v(c) := \exp\big(\hat{y}_i^v(c)\big)$, and $r_i^v(c) := \frac{\hat{P}_i^v(c)}{P_i^v}$, where P_i^v is the observed sale price.

From the validation predictions on fold v, we compute a predictive performance error score $\mathrm{E}_v(c)$ (e.g., mean-squared-error), together with a collection of fold-level assessment diagnostics $Q_{m,v}(c)$ indexed by m (e.g., PRD, VEI, and COD).

To aggregate results across folds, let $\alpha_v \geq 0$ satisfy $\sum_{v \in \mathcal{V}} \alpha_v = 1$, and define

$$\bar{\mathrm{Perf}}(c) := \sum_{v \in \mathcal{V}} \alpha_v \, \mathrm{Perf}_v(c), \qquad \bar{Q}_m(c) := \sum_{v \in \mathcal{V}} \alpha_v \, Q_{m,v}(c).$$

The default choice is uniform weighting, $\alpha_v = 1/V$, although the same formulation also allows non-uniform weights when one wishes to emphasize, for example, more recent validation windows.

Model Selection Formulation. We select a single deployed model by solving

$$\begin{aligned} c^\star \in \arg\max_{c \in \mathcal{C}} \quad & \bar{\mathrm{Perf}}(c) \\ \text{s.t.} \quad & \underline{\kappa}_m \leq \bar{Q}_m(c) \leq \overline{\kappa}_m, \qquad \forall m \in \mathcal{M}_{\mathrm{int}}, \end{aligned} \tag{33}$$

where $\mathcal{M}$ indexes diagnostics constrained to lie in admissible intervals.

Formulation (33) avoids collapsing the validation criteria into a one-dimensional weighted score: predictive performance is the objective, while ratio-study, regressivity, and dispersion diagnostics enter as feasibility constraints. This is the direct analogue of the compliance logic assessors already apply to candidate models, under which the selected model is the one with the best predictive performance among those that satisfy the chosen ratio-study tolerances.

The formulation above uses aggregated fold-level diagnostics as the default choice. When temporal stability is especially important, one may replace some aggregated constraints with foldwise requirements of the form

$$Q_{m,v}(c) \in [\underline{\kappa}_m, \overline{\kappa}_m] \qquad \forall v \in \mathcal{V}, \forall m \in \mathcal{M}$$

or, more generally, use a more conservative fold aggregator. We keep the weighted-average formulation as the main presentation because it yields a transparent calibration problem over a finite candidate set and aligns naturally with standard validation practice.

4.7.2 Alternative Pareto Criteria and a Stacking Extension

The constrained formulation in (33) has three known limitations: the constraints are enforced on validation data and need not hold out-of-sample, they may be jointly infeasible over the candidate set, and the predictive cost of tight tolerances can be large. To cover these cases we also consider two Pareto-based selection rules that do not require explicit tolerances: a *closest-to-Utopian* rule that picks the configuration closest (after normalization) to the ideal of the best attained value on each metric, and a *Nash equilibrium / maximum-volume* rule that picks the configuration whose aggregated-metric vector sacrifices proportionally the least on each dimension. Figure 5 illustrates the three rules on a two-metric example. We treat these rules as supplements to (33) rather than as coequal contributions; both are fully specified, with the convex-stacking extension, in Appendix A.

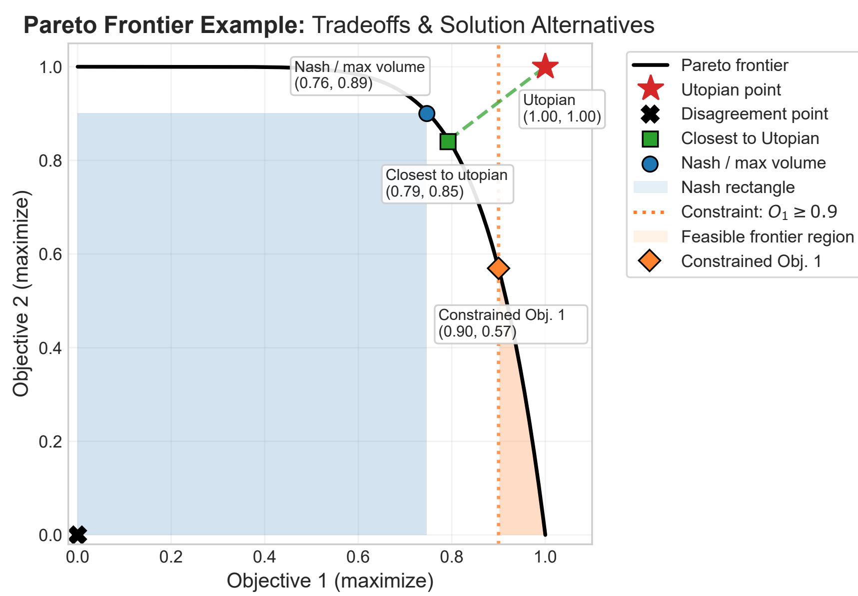

Figure 5: The three Pareto-optimal solutions on a two-objective convex frontier: constrained, closest-to-Utopian, and Nash / maximum-volume. The constrained rule is threshold-dependent; the other two do not require explicit tolerances.

Stacking extension. A natural further extension of (33) replaces discrete candidate selection by a convex combination of candidate predictions, which preserves the objective–constraint logic and can only enlarge the attainable accuracy–equity frontier. Because the assessor diagnostics are defined in terms of price-scale ratios, stacking is carried out on the price scale, and several ratio-study constraints admit an exact linear reformulation. The full stacking formulation and its convex-quadratic special case are deferred to Appendix A.

4.7.3 Empirical Calibration Results

Table 7 reports the held-out performance of models selected under different subsets of metrics. The constrained rule uses the intervals in Table 1. All selected configurations substantially improve regressivity, but the constrained versions become infeasible once non-regressivity metrics are added to the constraint set. This is consistent with the fact that our penalties target value-dependent trends rather than dispersion or central-tendency diagnostics, and it suggests that extending the penalty family to cover additional assessor-facing quantities is a natural next step.

Table 7: Held-out test set performance. The penalty parameter ρ is specific to the penalty definition.

Solution	Selected model	ρ	R^2	MAE	Med. r	Mean r	COD	COV	PRD	PRB	VEI
Unconstrained baselines											
BASELINE	Linear Reg.	–	0.762	$86,244	1.001	1.050	25.01%	36.86%	1.039	-0.011	-13.53%
BASELINE	LGBM	–	0.885	$71,912	0.944	1.021	23.07%	35.48%	1.085	-0.118	-37.64%
Constraints: PRD											
CONSTRAINED	LGBSurrogate	5.96	0.881	$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
UTOPIA	LGBSurrogate	100.00	0.877	$79,217	0.930	0.953	24.47%	34.89%	1.019	0.032	-4.41%
NASH	LGBSurrogate	2.95	0.883	$73,395	0.938	0.988	22.63%	33.63%	1.048	-0.034	-12.20%
Constraints: PRD, PRB, and VEI											
CONSTRAINED	LGBSurrogate	5.96	0.881	$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
UTOPIA	LGBSurrogate	7.91	0.882	$74,939	0.935	0.974	23.02%	33.85%	1.036	-0.007	-8.36%
NASH	LGBSurrogate	5.96	0.881	$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
Constraints: PRD, Mean r, and COD											
INFEASIBLE*	—	—	—	—	—	—	—	—	—	—	—
UTOPIA	LGBSurrogate	6.87	0.882	$74,449	0.937	0.977	22.84%	33.72%	1.038	-0.011	-9.38%
NASH	LGBSurrogate	2.95	0.883	$73,395	0.938	0.988	22.63%	33.63%	1.048	-0.034	-12.20%
Constraints: Full metric set†											
INFEASIBLE*	—	—	—	—	—	—	—	—	—	—	—
UTOPIA	LGBSurrogate	5.96	0.881	$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
NASH	LGBSurrogate	5.96	0.881	$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%

*INFEASIBLE abbreviates CONSTRAINED INFEASIBLE; no feasible selected solution.

† PRD, PRB, VEI, COD, COV, Mean r, Median r, and Weighted Mean r.

4.8 Practical Implications for Assessor Workflows

The empirical message is operationally simple. The proposed correction can be inserted into an existing assessor workflow with minimal disruption. It uses the same preprocessing pipeline, the same base model family, and the same downstream diagnostics already familiar to assessors. The only new tuning dimension is the penalty parameter ρ, whose qualitative effect is transparent: larger values tend to reduce value-dependent ratio trends. In that sense, the contribution is not a replacement for current workflows, but a targeted correction layer that can expand the attainable accuracy–equity tradeoff set while remaining easy to calibrate, audit, and communicate.

5 Discussion and Conclusions

We studied how to reduce regressivity in ML-based mass appraisal with minimal modification to existing assessor workflows. The main message is empirical: predictive accuracy alone does not guarantee vertical equity, and the move from linear regression to a nonlinear learner inside the CCAO pipeline improves R^2, MAE, COD, and COV while simultaneously worsening PRD, PRB, and VEI. A soft penalty on the covariance between log-residuals and log-prices recovers most of this equity gap at little cost in predictive fit, and does so with a single hyperparameter. The two penalty forms play complementary roles: the direct squared-covariance penalty is the closest instantiation of the dependence-control target; the separable surrogate is sample-additive and fits inside standard boosting custom-objective interfaces. In our CCAO runs the surrogate delivers the strongest combined performance and is the variant we recommend for routine deployment.

A second contribution is a calibration perspective. Once equity enters the loss through ρ, a single accuracy score no longer summarizes candidate quality, and the constrained calibration problem we use is a direct analogue of the ratio-study compliance logic that assessors already apply to candidate models. The alternative Pareto criteria of Section 4.7.2 and the convex-stacking extension described in Appendix A are available when explicit tolerances are unavailable or jointly infeasible; they did not dominate the constrained rule in our experiments, and we keep them as secondary tools rather than as coequal contributions.

Limitations. The method controls the covariance between log-residuals and log-prices, which is directionally aligned with the standard vertical-equity diagnostics but is not equal to them. In-sample or validation control of $\widehat{\mathrm{Cov}}(e, y)$ therefore does not imply out-of-sample PRD/PRB/VEI compliance, and we do not claim such a guarantee. The covariance is a first-order summary, so nonmonotone local distortions can cancel; the surrogate partially mitigates this by penalizing ratio deviations pointwise, but does not eliminate it. The penalties are global across the sample and do not currently use sub-market or geographic structure. The constrained calibration rule can become infeasible when many ratio-study tolerances are enforced simultaneously. Finally, the CCAO pipeline on which we evaluate the method is a translation rather than a direct execution of the production code, so the absolute levels we report should be interpreted within that translated pipeline.

Extensions. Two directions are especially natural. First, moving beyond first-order dependence via nonlinear measures such as HSIC or maximal correlation [27, 26], while keeping the separable implementation pattern used here. Second, stratifying the penalty by sub-market, neighborhood, or property class so that local value-dependent patterns are not averaged away by a global correction. Broader directions include applying the framework to additional assessor offices and base learners (random forests, neural networks, tabular foundation models), extending the constraint set to include central-tendency and dispersion diagnostics, and combining the penalty with interpretability tools that communicate the effect of ρ to non-technical audiences.

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A Alternative Pareto Criteria and Convex Stacking

This appendix collects the material for the two Pareto-based selection rules of Section 4.7.2 and for the convex-stacking extension of the constrained calibration problem. Both are deferred from the main text because they are secondary to the constrained rule in (33), which is the deployment rule we recommend.

A.1 Pareto-Based Selection Rules

Let $\mathcal{C}$ denote the candidate set, and let $\bar{\mathrm{Perf}}(c)$ and $\{\bar{Q}_m(c)\}_{m\in\mathcal{M}}$ denote the aggregated performance and diagnostic scores defined in Section 4.7. Combining the metrics into a single tuple $\bar{u}(c) := \big(\bar{\mathrm{Perf}}(c), \bar{Q}_1(c), \ldots, \bar{Q}_{|\mathcal{M}|}(c)\big)$, and applying a monotone rescaling into the unit box $[0,1]^{|\mathcal{M}|+1}$ in which larger is better on every axis yields a normalized score vector $\tilde{u}(c)$. The two rules below are defined on these normalized scores.

Closest-to-Utopian. Let the Utopian reference be $u^\star := \big(\max_{c\in\mathcal{C}} \tilde{u}_1(c), \ldots, \max_{c\in\mathcal{C}} \tilde{u}_{|\mathcal{M}|+1}(c)\big)$. The rule selects $c^\star_{\mathrm{Uto}} \in \arg\min_{c\in\mathcal{C}} \|u^\star - \tilde{u}(c)\|_2$.

Nash equilibrium / maximum-volume. The rule selects $c^\star_{\mathrm{Nash}} \in \arg\max_{c\in\mathcal{C}} \prod_k \tilde{u}_k(c)$, i.e., the configuration whose normalized score vector has the largest product across dimensions. Because the criterion is proportional, only the sign of each dimension matters.

Both rules operate on the same aggregated scores as (33) and therefore inherit the same fold structure. They differ only in how points on the empirical Pareto frontier are compared. In our CCAO study, the constrained rule, the closest-to-Utopian rule, and the Nash rule all select penalized surrogate configurations in the same region of the frontier (Table 7).

A.2 Convex Stacking Extension

A natural extension of (33) replaces discrete candidate selection by a convex combination of candidate predictions. Let $\mathcal{C}_{\mathrm{stack}} = \{c_1, \ldots, c_M\} \subseteq \mathcal{C}$ denote the stacking pool. For each fold v and observation $i \in V_v$, let $\hat{P}^v_{i,j} := \hat{P}^v_i(c_j)$ denote the out-of-fold price-scale prediction produced by configuration c_j. For weights w on the simplex $\Delta_M := \{w \in \mathbb{R}^M_+ : \sum_j w_j = 1\}$, define the stacked prediction on fold v:

$$\hat{P}^v_i(w) := \sum_{j=1}^{M} w_j\, \hat{P}^v_{i,j}, \qquad r^v_i(w) := \frac{\hat{P}^v_i(w)}{P^v_i}. \tag{34}$$

Stacking is defined on the price scale because the assessor diagnostics are ratio-study quantities on that scale. Fold-level performance and diagnostics $\mathrm{Perf}_v(w)$ and $Q_{m,v}(w)$ are computed exactly as in the single-model case, now as functions of w. The stacked calibration problem is

$$\begin{aligned} w^\star \in \arg\max_{w\in\Delta_M} \quad & \sum_{v\in\mathcal{V}} \alpha_v\, \mathrm{Perf}_v(w) \\ \text{s.t.} \quad & \underline{\kappa}_m \le \sum_{v\in\mathcal{V}} \alpha_v\, Q_{m,v}(w) \le \overline{\kappa}_m, \qquad m \in \mathcal{M}_{\mathrm{int}}. \end{aligned} \tag{35}$$

Once $w^\star$ is identified, the deployed stacked predictor is obtained by refitting each $c_j \in \mathcal{C}_{\mathrm{stack}}$ on the full pre-assessment development sample and combining its price-scale predictions with weights $w^\star$.

A convex quadratic special case. Take the average validation MSE as the objective and worst-fold box constraints on selected ratio-study diagnostics. Let $n_v := |V_v|$ and define the stacked MSE $\mathrm{MSE}_v(w) :=$

$n_v^{-1} \sum_{i \in V_v} \left(\hat{P}_i^v(w) - P_i^v\right)^2$. Consider the problem

$$\begin{aligned}
\min_{w \in \Delta_M} \quad & \frac{1}{V} \sum_{v \in \mathcal{V}} \mathrm{MSE}_v(w) && (36)\\
\text{s.t.} \quad & \underline{\tau}_{\text{mean}} \leq \frac{1}{n_v} \sum_{i \in V_v} \frac{\hat{P}_i^v(w)}{P_i^v} \leq \overline{\tau}_{\text{mean}}, \qquad \forall v \in \mathcal{V}, \\
& \underline{\tau}_{\text{wmean}} \leq \frac{\sum_{i \in V_v} \hat{P}_i^v(w)}{\sum_{i \in V_v} P_i^v} \leq \overline{\tau}_{\text{wmean}}, \qquad \forall v \in \mathcal{V}, \\
& \underline{\tau}_{\text{PRD}} \leq \frac{\frac{1}{n_v} \sum_{i \in V_v} \hat{P}_i^v(w)/P_i^v}{\sum_{i \in V_v} \hat{P}_i^v(w) \big/ \sum_{i \in V_v} P_i^v} \leq \overline{\tau}_{\text{PRD}}, \qquad \forall v \in \mathcal{V}.
\end{aligned}$$

The objective is convex quadratic in w because each $\hat{P}_i^v(w)$ is affine in w. The mean-ratio and weighted-mean-ratio constraints are linear in w. The PRD box constraints can be rewritten as linear inequalities on each fold:

$$\begin{aligned}
& \frac{1}{n_v} \sum_{i \in V_v} \frac{\hat{P}_i^v(w)}{P_i^v} - \underline{\tau}_{\text{PRD}} \frac{\sum_{i \in V_v} \hat{P}_i^v(w)}{\sum_{i \in V_v} P_i^v} \geq 0, \qquad \forall v \in \mathcal{V}, && (37)\\
& \frac{1}{n_v} \sum_{i \in V_v} \frac{\hat{P}_i^v(w)}{P_i^v} - \overline{\tau}_{\text{PRD}} \frac{\sum_{i \in V_v} \hat{P}_i^v(w)}{\sum_{i \in V_v} P_i^v} \leq 0, \qquad \forall v \in \mathcal{V},
\end{aligned}$$

so (36) is a convex quadratic program over the simplex.

B Algorithmic Details for Covariance-Penalized Boosting

This appendix collects pseudocode for the boosted-tree implementation of the squared covariance penalty introduced in Section 3.3. Throughout, we use the manuscript's sign convention

$$e_i = f(x_i) - y_i, \qquad c_i = y_i - \bar{y}, \qquad z = \widehat{\mathrm{Cov}}_n(e, y) = \frac{1}{n} \sum_{i=1}^{n} e_i c_i.$$

Under this convention, the squared covariance penalty is $\Psi^{\mathrm{cov}^2}(f) = \frac{1}{2} z^2$.

B.1 Exact Newton-style tree boosting with squared covariance penalty

The following algorithm implements the exact leaf-wise Newton update discussed in Section 3.4.2.

Algorithm 1 Exact Newton-style tree boosting with squared covariance penalty

Require: Training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, penalty weight $\rho \geq 0$, leaf regularization $\lambda \geq 0$, tree complexity penalty $\gamma \geq 0$, learning rate $\nu \in (0, 1]$, number of rounds M

Compute $\bar{y} \leftarrow \frac{1}{n} \sum_{i=1}^n y_i$
Compute $c_i \leftarrow y_i - \bar{y}$ for all $i = 1, \dots, n$
Initialize $f_0(x) \leftarrow \bar{y}$
for $m = 1, \dots, M$ **do**
  Compute current predictions $f_i \leftarrow f_{m-1}(x_i)$ for all $i = 1, \dots, n$
  Compute residuals $e_i \leftarrow f_i - y_i$ for all $i = 1, \dots, n$
  Compute covariance statistic

$$z \leftarrow \frac{1}{n} \sum_{i=1}^{n} e_i c_i$$

  Compute exact gradients

$$g_i \leftarrow \frac{e_i}{n} + \frac{\rho}{n} z c_i \qquad \forall i = 1, \dots, n$$

  Search over candidate tree structures T
  for each candidate tree T with leaves $I_1, \dots, I_J$ **do**
    for $j = 1, \dots, J$ **do**
      $G_j \leftarrow \sum_{i \in I_j} g_i$
      $n_j \leftarrow |I_j|$
      $s_j \leftarrow \sum_{i \in I_j} c_i$
    end for
    Form

$$D \leftarrow \operatorname{diag}\Big(\frac{n_1}{n} + \lambda, \dots, \frac{n_J}{n} + \lambda\Big), \qquad \beta \leftarrow \frac{\rho}{n^2}, \qquad A_T \leftarrow D + \beta s s^\top$$

    Compute exact tree gain

$$\mathrm{Gain}(T) \leftarrow \frac{1}{2} G^\top A_T^{-1} G - \gamma J$$

  end for
  Select the best tree structure T_m
  For the selected tree, form $A_m \leftarrow D + \beta s s^\top$
  Compute optimal leaf weights

$$w^\star \leftarrow -A_m^{-1} G$$

  Update

$$f_m(\cdot) \leftarrow f_{m-1}(\cdot) + \nu \, T_m(\cdot; w^\star)$$

end for
return final predictor f_M